

Domain Classification WSD systems

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Outline

- Domain classification
 - System
 - Evaluation
- Word Sense Disambiguation
 - Systems
 - timbl-DSC
 - svm-DSC
 - ukb-DSC
 - Evaluation
 - Fold-cross
 - Random
 - All-words

Domain classifier

- Automatic system to assign ***domains labels*** to texts
- 37 domains created by grouping WordNet Domains
 - Biol -> anatomy, biology, botany, ecology, entomology, genetics, zoology and physiology
- Support vector machines (SVMLight, Joachims 1998)
 - One binary classifier per domain
- Features:
 - Bag-of-words approach (binary features)

Domain classifier

- Training data:
 - Synonyms and definitions from Cornetto synsets tagged with domains
- Evaluation test sets:
 - Random_set: 143 paragraphs randomly selected from the 1st and 2nd release of SONAR
 - Random_genre_set: 170 paragraphs, where we took randomly a few from each genre in the 1st and 2nd release of SONAR
 - Manually annotated with domains

Domain classifier

- A paragraph is considered if:
 - At least **one** of his related domains is returned by the classifier within the ***top 5 scoring documents***
- Accuracy, ok / (ok+wrong)
 - Random_set 84.62 %
 - Random_genre_set 79.88 %
- All SONAR paragraphs have been automatically assigned with their domains
 - 9.4 M of paragraphs in SONAR annotated

WSD systems: timbl-DSC

- Based on TiMBL, supervised K-nearest neighbor classifier (Daelemans et al, 2007)
- One classifier per word (multi-class classification)
- Memory-based learning
 - All trained instances are stored along with the senses associated
 - To tag a new example:
 - Find the 'k' most similar examples in the stored model
 - Return the majority sense of these 'k' examples

WSD systems: timbl-DSC

- Features
 - Local context: words, lemmas, PoS in context
 - Global context: filtered bag-of-words (min 5, 0.8 relative frequency)
 - Domain information:
 - Sonar category
 - Domain labels
- Timbl parameters
 - Value for 'k', algorithm and feature metric, weighting scheme...
 - Optimization per classifier: leave-one-out

WSD systems: svm-DSC

- Based on Support Vector Machines (SVMLight, Joachims 1998)
- Supervised binary linear classifier
 - Represent all training instances in a n-dimensional space (most simple 2D)
 - Learn a line that separates both sets of examples
 - Maximize the margin of separation of the line with the two groups of examples
 - To classify a new instance:
 - Represent it on the 2D space and see in which side of the line falls

WSD systems: svm-DSC

- One classifier per word
 - SVMLight is binary in principle
 - *One-vs-all*: one binary classifier per word sense
 - Positive examples of the sense
 - Negative examples of the rest of senses
- Features:
 - Bag of words
 - Filtering by relative frequency per classifier
- Default svm parameters mostly used in WSD systems

WSD systems: ukb-DSC

- Knowledge-based system (unsupervised) (Agirre and Soroa 2009)
- WordNet (Cornetto) is considered as a graph where:
 - Synsets: nodes
 - Relations: edges
- Personalized PageRank algorithm
 - Modification of PageRank
 - Context words act as source nodes injecting mass into word senses
 - Assign stronger probabilities to certain nodes

WSD systems: ukb-DSC

- Dutch WordNet
- English WordNet
- Dutch WordNet ==> English WordNet
- WordNet Domain
 - *tennis player, tennis ball => tennis ==>*
 - *Football player, football => soccer =*
- Annotation co-occurrence relations
 - *Polysemous ==> monosemous*
 - *Polysemous ==> polysemous*



SPORT

WSD Systems

- Three systems
 - 2 supervised systems
 - timbl-DSC
 - svm-DSC
 - 1 unsupervised system
 - ukb-DSC
- One *super-system* combining the 3 systems
 - Majority voting
 - We have tried different weights for each system (decide in case of tie)

WSD. Evaluation

- We have a huge amount of evaluation results
 - Three systems (and combination) with different configurations for each
 - Three types of evaluation
 - Separate results for nouns, verbs and adjectives
 - Results for systems, lemmas and word meanings
 - For senses (lexical units), sense-groups and base concepts
- All the results and evaluation data is available on the **website**
- In this presentation: ***best overall results for senses***

WSD. Evaluation

- Three different evaluations (each one with a specific goal)
 - Fold cross validation
 - To get the *best sense tagger* on SONAR, to fulfill the main goal of the project
 - Random evaluation on SONAR
 - To *estimate the accuracy* of the sense tagger over the *rest of SONAR*
 - All words evaluation
 - To analyze the *performance* of our SONAR-oriented WSD system in *totally independent texts*

WSD. FC Evaluation

- Token accuracy for systems. Senses
 - Using manually annotated data of the AL process

		Nouns	Verbs	Adjectives
timbl-DSC	No domain feats.	83.97	83.44	78.64
	<i>Domain features</i>	<i>81.60</i>	<i>81.21</i>	<i>76.28</i>
svm-DSC	No domains feats.	81.17	84.19	77.88
	<i>Domain features</i>	<i>82.69</i>	84.93	79.03
ukb-DSC	UKB4f (all relations 1,7M relations)	73.04	55.84	56.36
	UKB5d (no singletons 1,1M relations)	51.29	37.52	37.78
	UKB1 (cornetto + domain relations 138,427)	47.03	30.61	35.36

WSD. FC Evaluation

- Token accuracy for the combination of the systems. Senses

	timbl-DSC	svm-DSC	ukb-DSC	Token accuracy
Nouns	1			83.97
	1.5	1	1	88.53
	1	1	1.5	88.65
Verbs		1		84.93
	1	1.5	1	87.60
Adjectives		1		79.03
	1	1.5	1	82.97
	1.5	1	1	83.06

WSD. Random Evaluation

- Token accuracy for the random evaluation. Senses
 - 5 nouns, 5 verbs and 3 verbs selected from:
 - Between 90 and 100 (not with acc=100)
 - Between 80 and 90
 - Between 70 and 80
 - Between 60 and 70

System	<i>Nouns</i>	<i>Verbs</i>	<i>Adjectives</i>
timbl-DSC	54.25	48.25	46.50
svm-DSC	64.10	52.20	52.00
ukb-DSC	49.37	44.15	38.13
<i>Combination 1 – 1.5 - 1</i>	66.92	60.55	55.11

WSD. All words

Token_id no available as feature

- Around 8 points of decrease if not used in FC validation

`<w xml:id="WR-P-P-G-0000148955.p.28.s.3.w.8"><t>paard</t>...`

--> *Sense of chess piece*

....

`<w xml:id="WR-P-P-G-0000148955.p.30.s.3.w.5"><t>paard</t>`

--> *Sense ??? Maybe chess piece??*

System	Nouns	Verbs	Adjectives
timbl-DSC	55.76	37.96	49.09
svm-DSC	64.58	45.81	55.70
ukb-DSC	56.81	31.37	35.93
Combination 1 – 1.5 - 1	66.09	45.68	52.24

WSD. Overall Evaluation

Overall results for systems, considering *all the lemmas*

System	Nouns	Verbs	Adjectives
Fold cross validation	88.65	87.60	83.06
Random evaluation	66.92	60.55	55.11
All words evaluation	66.09	45.68	52.24

WSD. Overall Evaluation

- Distribution of word in terms of performance for the combine system in the fold-cross validation
 - Performance for ***each lemma***

Range	Nouns	Verbs	Adjectives	All
P >= 80	87.91 %	91.03 %	69.55 %	82.83 %
70 <= P < 80	9.86 %	7.35 %	22.69 %	13.30 %
60 <=P < 70	2.08 %	1.18 %	6.57 %	3.28 %
P < 60	0.15 %	0.45 %	1.19 %	0.60 %

Sense tagging of all SONAR

- We apply our three systems to all unannotated SONAR (for the DSC selected words)

Number of tokens automatically annotated:

	Nouns	Verbs	Adjectives
timbl-DSC	18,5 M	23,9 M	5,3 M
svm-DSC	18,5 M	23,9 M	5,3 M
ukb-DSC	18,9 M	24,1 M	5,4 M

- All automatic annotations with confidence value
 - You can select the best ones

Sense tagging of all SONAR

Number of tokens automatically annotated with a ***confidence ≥ 0.8***

	Nouns	Verbs	Adjectives
Timbl-DSC all	18,5 M	23,9 M	5,3 M
Timbl-DSC Conf≥ 0.8	10,8 M (58%)	15,6 M (64%)	2,6 M (50%)

Around **29 M** tokens with a confidence ≥ 0.8

**Dank je wel !
Thanks !
Gracias !**

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